

Two Preregistered Nulls: The Loud Commitment vs the Silent One, and the Constraint Geometry That Wasn't

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Cross-arc joint synthesis **Date:** 2026-07-27 **Scope:** DCR3 (DR-arc textual nominator on real material) + Constraint-Swap Causal Geometry (32-seed representational-deformation study, run independently by the human director).

Abstract

Two independently designed load-bearing experiments were preregistered and executed on 2026-07-27. Each targeted the strongest claim its research arc could make on a case where the answer could be checked. Each hit its stated gate. Each failed cleanly, in a way that was predicted-in-principle by the arc's own theoretical work but not yet demonstrated on real material.

DCR3 asked whether the DR-arc's execution-free scoring function ranks Einstein's actual 1905 deletion (the T1 = absolute-simultaneity class) first on the pre-1905 electrodynamics corpus. Result: T1 came in **third**, behind unclassified (rank 1) and T2_privileged_frame (rank 2). The scoring function was not broken – it correctly identified T2 as the most-defended commitment in the corpus, which historically it was. But **the most-defended commitment is not the deletable one**. Silent commitments are deletable precisely because nobody is defending them, and Einstein's move required exactly that asymmetry.

Constraint-Swap Causal Geometry asked whether a task constraint that determines successful-future reachability causally deforms a recurrent agent's hidden geometry, and whether targeted low-rank transports of that geometry selectively reverse or accelerate the corresponding behavior. Result: 32 seeds, all preregistered non-compensatory gates G1–G5 failed. The agents were competent (mature A/B/D accuracy = 1.000), the sham stayed at chance (0.509), the injected-geometry recovery lift (0.907) confirmed the intervention machinery works. But **constraint-specific reachability geometry did not appear, swaps did not track, and rank-4 affine transports did not selectively drive behavior**. In a controlled regime with exactly enumerable reachable futures, the constraint→ geometry→ behavior causal chain did not hold.

These are the honest results. They are also the strongest form of each arc's claim that could be tested in a session with proper discipline. Both fail. The failures constrain what the next attempt would need to be.

1. What each experiment was designed to establish

DCR3. Preregistered scoring: $\text{score}(p) = \text{kind_weight}(p) \times \text{degree}(p)$. Class-level sum with $\text{multidoc}(\text{min_docs}=2)$ gating. Ground truth: T1 class per DCR1c/d/e/f/2a/2b. Baseline: 10,000 random permutations. Four non-compensatory gates (M1 T1-first-at-1904, M2 T1-not-first-at-1880, M3 beats-random-null, M4 scoring-committed). The intended-positive outcome was the strongest empirical statement the DR-arc could make: *"execution-free algorithmic nomination identifies the commitment a scientific community deleted, on a real case with ground truth."* First-of-its-kind if it landed.

Constraint Swap. Preregistered by the human director on 2026-07-27 03:41 EDT before implementation. Corrected once (2026-07-27 03:48) after an F0 unit test rejected the initial vertical/horizontal-fiber design as unable to distinguish metric deformation from coordinate rename. Active design: 6×6 torus (7×7 horizontal cylinder for topology transfer) with two orthogonal even-parity constraints (checkerboard C_A, horizontal-stripe C_B), a learnable

deterministic control C_D (vertical-stripe parity), and a randomised sham. Frozen meta-GRU agents. Five non-compensatory gates (G1 constraint-specific geometry, G2 swap tracking, G3 selective impairment, G4 selective rescue, G5 topology transport), each requiring both directions and preservation controls. The intended-positive outcome: "*constraint-conditioned reachability- aligned hidden subspace is causally relevant to behavior in this controlled regime.*"

Both are the strongest testable version of their arc's central claim. Both are ground-truthed. Both preregistered non-compensatory gates and honored the verdicts.

2. What each experiment established

DCR3 result (from PR #442)

The DR-arc nominator scored 1904 as:

rank	class	score	members	documents
1	unclassified	302	235	15
2	T2_privileged_frame	15	11	8
3	T1_absolute_simultaneity	9	7	5
4	T3_local_time_artifice	2	2	2

M1 NO_GO (T1 rank 3 $\neq$ 1). M2 GO. M3 NO_GO by construction. M4 GO. Overall NO_GO.

Substantive finding: T2 (aether frame) outscored T1 (simultaneity) because T2 was more widely stated and defended across the electrodynamics literature – exactly what the scoring rule measures. Einstein did not delete T2. He deleted T1, and T1 scored low because absolute simultaneity had gone silent in the physics community by 1900 (as DCR1d showed on Newton's Scholium). *Silent = undefended = deletable*. The DR-arc's assumption "load-bearing in text $\Rightarrow$ deletable in a revolution" runs backward on this case.

Constraint Swap result (from PREREGISTRATION + summary.md)

32 preregistered confirmatory seeds, 10,000 bootstrap resamples, active 6x6 torus + transfer 7x7 cylinder.

gate	verdict
F0 integrity + identifiability	PASS (registered component tests)
F1 competence + measurement sensitivity	PASS (A, B, D all 1.000; sham 0.509; injected-geometry recovery lift 0.907)
G1 constraint-specific geometry	FAIL (A-specific -0.363, B-specific -0.237)
G2 swap tracking	FAIL (A→B -0.594, B→A -0.593)
G3 selective impairment	FAIL (undo-B -0.187, undo-A -0.225)
G4 selective rescue	FAIL (impose-B -0.057, impose-A -0.014)
G5 topology transport	FAIL (all transfer sub-gates)

Decision: REJECT_CONSTRAINT_SPECIFIC_DEFORMATION.

The controls are the load-bearing part of the failure. F0 identifiability passed, F1 competence + measurement sensitivity passed, sham stayed at chance, and the positive-control geometry injection was recovered with lift 0.907. So the failure of G1–G5 is not "the intervention didn't work" or "the agent couldn't learn the task" – it is: **the agent solved the task, the interventions on constructed geometry worked, but the geometry the interventions were supposed to target does not appear to carry the constraint-specific information in the way the hypothesis predicted.** Under this implementation, in this controlled regime, the causal chain does not hold.

3. What the joint reading is

Both arcs' strongest testable claims failed in exactly the direction each arc's theoretical machinery said they might.

distinguish a commitment $\$D\$$ from any specific realisation when $\$D\$$ has multiple non-equivalent surface forms. DCR3 confirms this on real material: T1's 6-register spectrum (DCR1f) is distributed across the corpus in a way that surface-based scoring systematically underweights.

canonical form hit a graded wall whose severity depends on domain-general proxies. Constraint Swap tested a version of this in a different modality: an agent's *internal* representation under task-constraint variation. The result is even sharper – no proxy signal was found either. The active reachability geometry did not track the active constraint, even in a controlled world where reachability was exactly enumerable.

- **DR5 / DR7** predicted that proposition-ranking nominators cannot
- **DR5\|\|**** predicted that verifiers with weak access to $\$D\$$'s

The two arcs together map a **specific failure surface**: on real conceptual-change material with ground truth (DCR3) *and* in a controlled agent-representation regime with ground truth (Constraint Swap), the strongest currently-testable versions of both "algorithmic identification of the target deletion" and "causal constraint→geometry→behavior chain" do not hold.

This is not "nothing works." It is *"here are two specific architectural approaches that were tested with proper discipline and did not clear their load-bearing gates."* Both results are publishable-shaped, both are informative, and both constrain what the next serious attempt has to look like.

4. What survives

computational-verification questions CAN be operationalised with preregistered gates, single-shot verdicts, and honest reporting of the null. DCR3 followed DCR1's discipline (preregistration written first, scoring function SHA-256-pinned). Constraint Swap followed its own more stringent discipline (F0 unit tests rejected the first design *before any training*; the pivot to balanced parity-based constraints was recorded as an amendment).

triangulation now stand as a coherent theoretical framework whose empirical predictions have been tested twice, with both tests producing the predicted class of null. That is not "the framework is wrong" – that is "the framework correctly predicted the shape of the wall in advance."

(defended commitments are protected, silent commitments are deletable) is a substantive claim about how conceptual revolutions happen. It is testable on other cases (Copernicus, Darwin, Lavoisier – each with known deletions) and would extend the DR-arc's empirical scope if replicated.

- **The methodology.** Both arcs demonstrate that scientific
- **The characterisation.** DR5 + DR7 + DR5b + the DR6 empirical
- **The specific new finding from DCR3.** The loud/silent asymmetry

5. What does not survive

deletability.** DCR3 shows this is false in the direction that matters for revolutions.

possibility" grand principle, in its unmodified form.** Constraint Swap tested a specific empirical claim that would follow from the principle if it were true of learned recurrent agents in controlled worlds, and rejected the claim on 32 seeds. The principle may still be right in some broader sense – as the other agent noted, unifying principles are always at risk of surviving as evocative metaphors – but as a mechanistic claim it has been tested here and did not survive.

- **The DR-arc's assumption that textual load-bearing predicts
- **The "meaning is constraint-induced deformation of reachable

6. What comes next (both arcs' honest continuations)

DR-arc. Two paths:

score for "how presuppositional is this commitment given the corpus's arguments." DR6/DR6e showed LLM semantic access escapes DR5's wall for accessible $\$D$; this would test whether it reaches T1 on the electrodynamics corpus. Not a proposition-ranking nominator any more – the LLM does the ranking. Preregister the scoring prompt before running.

scorer based on corpus statistics cannot identify silent-but-presupposed commitments. Would generalise DCR3's empirical finding into a specific corollary of DR5/DR7.

- **Semantic-scoring DCR3b.** Replace degree with a Claude-based
- **Theorem-shape DR8.** Prove formally that any proposition-ranking

Constraint-swap arc. Two paths:

toy meta-GRU on a 6×6 torus. The same experiment on a real vision-language model with activation-patching interventions would be a much larger test. Predicted based on the toy result: probably still fails, but the failure would be in a regime that interpretability researchers already care about.

intervention gates (G3–G5) and just ask whether the geometry *correlates* with the constraint. Constraint Swap's own metrics (G1 A-specific -0.363) suggest the answer is no even at this weaker level – but a re-designed metric might catch a residual correlation. This would be a descriptive claim, not the mechanistic claim the original was aimed at.

- **Real-network Constraint Swap.** The 32-seed rejection was in a
- **Weaker "descriptive geometry" claim.** Drop the causal

7. Honest answer to "is this major?"

No, and now I have a specific reason to say so beyond the earlier "not-yet" framing.

Both load-bearing tests failed. That is not "major-and-negative" in the sense of a paradigm-shifting refutation of a widely-held claim. It is: two well-scoped attempts hit their scoped limits. The DR-arc's textual approach does not reach the deletion Einstein made; the constraint-geometry approach does not establish the causal chain in a controlled toy regime.

What this session's work IS, if honestly labelled:

computational-scientific claims that would otherwise slip.

two domains and two verifier architectures.

independent research programs' central claims.

that would extend if replicated.

- A working methodology for preregistered verification of
- Two theorems (DR5, DR7) with an empirical triangulation across
- Two clean nulls on the strongest load-bearing versions of two
- One substantive empirical claim (the DCR3 loud/silent asymmetry)

That is a real and honest contribution. It is not major in the "changes what the field thinks" sense. Being clear about what a result IS is more useful than inflating it.

Appendix: reproduction

DCR3:

```
uv run --no-sync python -m experiments.date_cut_retrodiction.run_dcr3
```

Constraint Swap: See `experiments/constraint_swap_causal_geometry/README.md` for `run_experiment.py` invocation and 32-seed configuration.

Both experiments' preregistrations, gate verdicts, and raw data committed to the repository.
All prior arcs unchanged.